CHAPTER 1

INTRODUCTION

This chapter introduces the project *LifeLine Mini*, an offline, AI-enabled portable ECG monitoring and emergency alert system. The project aims to address the challenges of detecting cardiac abnormalities in remote, rural, and low-connectivity environments. This chapter discusses the background of the project, previous work in the field, the motivation behind developing the system, the core problem addressed, the objectives set, the proposed design approach, and the organization of the entire report.

Electrocardiography (ECG) is one of the most essential diagnostic tools used for monitoring heart activity. However, access to ECG equipment is still limited in many areas and emergency situations. Most existing systems rely heavily on smartphones, cloud servers, or continuous internet connectivity, which makes them unsuitable for rural environments and scenarios where timely medical assistance is unavailable. The proposed system aims to bridge this gap by offering a completely independent and user-friendly solution capable of recognizing abnormal cardiac activity and sending emergency alerts autonomously.

1.1 Overview

LifeLine Mini is a compact, battery-powered ECG monitoring device built using the ESP32 microcontroller, AD8232 ECG front-end, SIM800L GSM module, and NEO-7M GPS module. The system continuously reads ECG signals, processes them using an Exponential Moving Average (EMA) filter, and classifies them using a TinyML-based arrhythmia detection model deployed directly on the ESP32. If the model detects an abnormal heart rhythm, the device automatically sends an emergency SMS alert with real-time GPS coordinates to predefined contacts. The device includes a 0.96-inch OLED screen for displaying live ECG waveforms and status notifications, and a MicroSD card interface for storing ECG data for later medical review. It is designed to work entirely offline, making it highly reliable in real-world emergency conditions.

1.2 Previous Work

Many researchers and developers have proposed IoT-based and mobile-health ECG systems. Earlier systems typically used microcontrollers such as Arduino or Raspberry Pi for ECG acquisition and relied on Wi-Fi or Bluetooth for data transmission to smartphones or cloud servers. While these approaches improved accessibility, they still had critical limitations. Most systems required stable internet connectivity, cloud-based analysis, or a user-controlled smartphone. Other systems lacked autonomous emergency alerting or real-time arrhythmia detection on the device.

Several papers also explored wearable ECG devices, Bluetooth-enabled ECG streaming, cloud-based health dashboards, and smartphone-integrated ECG interpretation tools. Despite their contributions, these solutions fail to operate effectively when the user is unconscious, offline, or unable to manually activate the system. Additionally, they did not integrate GPS tracking or GSM-based emergency communication. These gaps highlight the need for a device that can function independently and provide immediate life-saving alerts without external dependencies.

1.3 Motivation

Sudden Cardiac Arrest (SCA) and similar heart-related emergencies occur unpredictably. Immediate detection and timely intervention are essential to improve survival rates. Millions of individuals living in rural areas, especially in developing nations, do not have access to ECG machines or nearby hospitals. Even in cities, emergencies can occur in situations where the victim is alone or unable to call for help.

The motivation of this project is to build a device that requires no internet, no smartphone, and no human intervention during emergencies. By enabling real-time ECG monitoring with autonomous detection and SMS alerting, the system can play a crucial role in saving lives. Its portability, low power consumption, and affordable hardware components make it feasible for large-scale use in communities, elderly care, remote health centers, disaster zones, and field medical camps.

1.4 Problem Statement

Most existing ECG monitoring systems depend on external infrastructure such as Wi-Fi, smartphones, cloud processing, or hospital equipment for diagnosis and emergency communication. These dependencies make them unreliable in remote locations, emergencies, or situations where the patient loses consciousness. These systems also lack built-in arrhythmia detection, GPS localization, and offline emergency alerting.

Therefore, the problem addressed in this project is:

"To design and develop a portable, offline-capable ECG monitoring system with on-device arrhythmia detection and automatic GSM-based emergency alerting with GPS location, suitable for rural and low-resource environments."

1.5 Objective of the project

The main objectives of this project are:

1. To design a portable ECG monitoring device using AD8232 and ESP32.
2. To implement a TinyML model for real-time arrhythmia detection directly on the microcontroller.
3. To integrate SIM800L GSM module for automatic SMS alerts during emergencies.
4. To incorporate NEO-7M GPS for real-time location tracking.
5. To store ECG data logs on a MicroSD card for future medical review.
6. To provide real-time ECG visualization using an OLED display.
7. To ensure the device operates completely offline and independently.
8. To develop a low-cost, power-efficient device suitable for field use.

1.6 Proposed System

The proposed system continuously captures ECG signals through the AD8232 sensor and processes them on the ESP32 microcontroller. The filtered ECG signals are analyzed using an embedded TinyML model trained to classify normal and abnormal heart rhythms. When an abnormal pattern is detected, the system retrieves GPS coordinates from the NEO-7M module and sends an SMS alert containing the location and status to emergency contacts via the SIM800L GSM module. The user interface includes an OLED screen showing ECG waveform and system parameters. The device stores time-stamped

ECG data on an SD card for later analysis. All components are powered by a rechargeable 18650 battery, enabling long hours of operation.

1.7 Organization of the report

This report being submitted is organized into five chapters.

CHAPTER 1 provides a brief introduction to project, including the background, motivation, problem statement, objectives, and the proposed system

CHAPTER 2 explains about the papers that we have surveyed to understand the existing technologies, research gaps, and the basis for selecting our proposed methodology.

CHAPTER 3 describes about the design methodologies that we have have chosen to implement the system, including hardware components, software workflow, block diagrams, and functional analysis.

CHAPTER 4 contains the result that we have obtained after the work done.

CHAPTER 5 gives conclusion and future scope of this project. It also contains advantages and applications of the proposed system.

CHAPTER 2

LITERATURE SURVEY

In this chapter, a comprehensive survey of existing research in the areas of ECG monitoring, IoT-based health systems, embedded biomedical devices, GSM/GPS emergency alerting, and TinyML-enabled diagnostics is presented. The goal of this survey is to understand current technological approaches, identify their limitations, and highlight the research gap that motivates the development of *LifeLine Mini*. The reviewed papers discuss a wide range of cardiac monitoring systems, including portable ECG devices, smartphone-integrated solutions, cloud-based health platforms, and embedded AI-driven diagnostic frameworks. Each study provides insights into specific methodologies, yet collectively reveals significant shortcomings such as internet dependency, lack of real-time autonomous detection, absence of emergency alerting, and limited support for offline operation. These gaps justify the need for a fully independent, portable, and intelligent ECG system capable of functioning in rural and emergency settings. By analyzing these existing solutions, the strengths and weaknesses of different design strategies become clear, enabling informed decisions for system optimization.

2.1 Survey Papers

Artificial Intelligence Enhanced Portable Electrocardiogram Monitor

Authors: Viji K. et al.

Publication: IEEE ICCCNP, 2025

This paper proposes a portable ECG system enhanced with unsupervised machine learning using the DBSCAN algorithm. The system captures ECG signals using an AD8232 sensor and processes them via ESP8266, visualizing results on a cloud dashboard with color-coded indicators. While it improves accessibility and provides AI-based classification, the method depends on Wi-Fi and cloud platforms for dashboard access and alerting. It does not include GPS tracking, GSM alerts, or offline analysis. These limitations emphasize the need for a system like *LifeLine Mini*, which works fully offline and supports automatic SMS alerting.

IoT-Based ECG Monitoring System on Smartphones

Authors: Mahmudul Hasan Abid, Iffat Ara Talin

Publication: IEEE iCACCESS, 2024

This system transmits ECG data via Bluetooth from an ESP32-AD8232 setup to a smartphone application, enabling real-time visualization. Validation with clinical ECG machines shows close waveform similarity. However, the system is entirely dependent on a functioning smartphone, Bluetooth connectivity, and user interaction. This makes it unreliable during emergencies when the patient is unconscious. The absence of GPS or autonomous alerts highlights the gap filled by *LifeLine Mini*, which removes smartphone dependency and operates autonomously.

Smart and Wearable ECG Monitoring System as a Point-of-Care Device

Authors: Abhinay Vishwanatham et al.

Publication: IEEE ANTS, 2018

The authors present a wearable ECG device using ADS1293 and BLE communication for real-time monitoring. The system uploads data to a server for physician review. While it offers portability and multi-lead accuracy, it relies on internet connectivity and smartphone pairing. It lacks offline classification and automated emergency alerting. *LifeLine Mini* addresses these gaps by introducing TinyML-based on-device detection and GSM-based emergency alerts without external infrastructure.

IoT Based Health Monitoring System Using Raspberry Pi – Review

Authors: Sunilkumar L. Rohit, Bharat V. Tank

Publication: IEEE ICICCT, 2018

This review discusses multiple IoT-based health systems using Raspberry Pi as a central gateway to collect vital signs and send data to cloud dashboards. While the approach is powerful for remote monitoring, it is unsuitable for emergency use due to high power consumption, cloud dependency, and the absence of autonomous detection. Unlike these systems, *LifeLine Mini* eliminates the need for cloud servers and performs all processing on-device. However, the reliance on continuous internet connectivity makes such systems unreliable in rural or low-network regions.

AI-Based Detection for Remote Electrocardiogram Monitoring System

Authors: Max Garcia, Sanjeev Kumar

Publication: IEEE AIIoT, 2024

This paper offers a cloud-connected ECG monitoring system using Arduino MKR, AD8232, and ML models like Random Forest trained on MIT-BIH datasets. The system uploads all data to a remote server for analysis. The study also explores cybersecurity threats and their impact on data accuracy. However, reliance on constant internet access and cloud-side ML limits emergency responsiveness. *LifeLine Mini* improves upon this by enabling on-device TinyML detection and offline emergency alerts.

The Design of Portable ECG Health Monitoring System

Authors: Bo Liu et al.

Publication: IEEE CCDC, 2017

This system uses the BMD101 ECG chip and an STM32 microcontroller to display ECG signals, store data on TF cards, and support USB/Bluetooth connectivity. It is effective as a portable device but lacks GSM alerts, GPS tracking, and autonomous detection. These missing features are directly addressed in *LifeLine Mini* through GSM-based alerting and real-time arrhythmia classification. Additionally, while the original design focuses mainly on signal acquisition, LifeLine Mini extends functionality by integrating intelligent on-device analysis and automated emergency response.

A Low-Cost IoT-Based Health Monitoring System

Authors: Samit Hasan, Muhammad Abdullah Arafat

Publication: IEEE ICECE, 2022

This low-cost system uses ESP32 with multiple biomedical sensors and uploads data to a cloud server via MQTT. While it is affordable and accurate, the need for Wi-Fi connectivity and cloud storage makes it unsuitable for emergencies. *LifeLine Mini* overcomes this limitation by functioning entirely offline and using GSM for critical alerts. Moreover, the dependence on cloud platforms introduces latency, which can delay critical decision-making during medical emergencies. LifeLine Mini avoids this delay by performing on-device TinyML inference, ensuring immediate classification and faster response times.

Portable ECG Monitoring System Design

Authors: Tang Lili, Huang Wei

Publication: IEEE, 2019

The system uses AD8232, MSP430, FIR filtering, SD card logging, and LCD display. Additionally, Wi-Fi transmission supports remote access. However, absence of AI-based analysis and GSM/GPS limits emergency functionality. The proposed system extends this work with embedded TinyML and offline alerts.

Smart Health IoT Monitoring System for Real-Time Arrhythmia Detection Using AI

Authors: S. Jalaja et al.

Publication: IEEE ICDSBS, 2025

A multi-sensor system with CNN-based arrhythmia detection, cloud integration, and ThingSpeak dashboard. Although highly accurate, it requires Wi-Fi and cloud platforms. No GSM/GPS is included. *LifeLine Mini* replaces cloud inference with on-device TinyML and offers offline emergency response.

IoT-Based Wireless Patient Monitor Using ESP32 Microcontroller

Authors: Mujeeb Rahman K.K. et al.

Publication: IEEE ACIT, 2023

Uses ESP32, AD8232, MAX30102, Firebase cloud, and Android apps for multi-parameter monitoring. Cloud dependency and the absence of autonomous emergency functionality make it unsuitable during connectivity failures. The proposed solution overcomes this by enabling GSM-SMS alerts and offline functioning.

An IoT Based Healthcare Using ECG

Authors: Rami Hodrob et al.

Publication: IEEE ACIT, 2023

This Raspberry Pi-based ECG system uploads real-time ECG data to the cloud and sends AFib alerts using smartphone GPS. While it includes SMS alerts, it still relies on cloud servers, smartphones, and high-power hardware. *LifeLine Mini* improves upon this by using a low-power embedded platform with built-in GPS and independent SMS alerting.

Summary of Literature Survey

Across all eleven papers, major limitations include dependence on internet connectivity, cloud servers, smartphones, and external devices for ECG processing or emergency alerting. Most systems do not provide on-device arrhythmia detection, autonomous emergency handling, or integrated GPS-based location tracking. These common shortcomings clearly indicate the need for a portable, offline, self-contained ECG emergency device—precisely what *LifeLine Mini* delivers through TinyML analysis on ESP32, GSM SMS alerting, and GPS integration.

CHAPTER 3

DESIGN METHODOLOGIES

In this chapter, the overall design methodology of *LifeLine Mini* is explained in detail. The system integrates biomedical signal acquisition, embedded processing, TinyML-based classification, GPS geolocation, GSM emergency alerting, and data logging into a compact two-layer device. The design methodology covers hardware architecture, software flow, signal processing pipeline, and the functional behavior of each module. The goal of the proposed methodology is to ensure that the system operates reliably in offline environments and provides real-time emergency support without external dependencies such as the internet or smartphones.

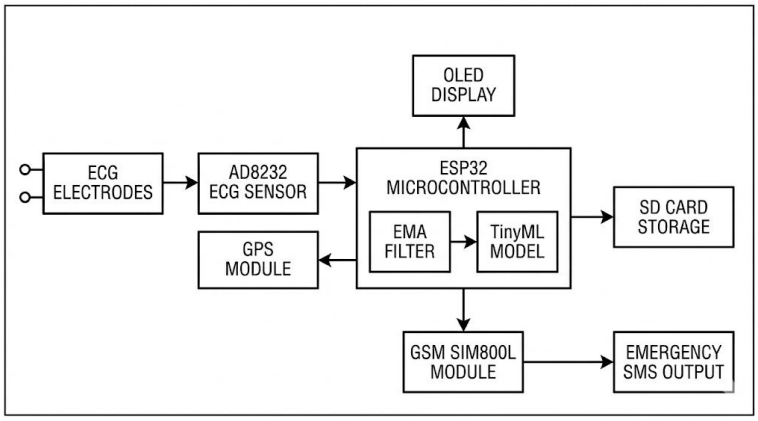

Fig. 3.1 Block Diagram of the Proposed ECG Monitoring System

The system begins by collecting raw ECG signals through electrodes connected to the **AD8232 ECG sensor**, which amplifies and filters the cardiac waveform. The cleaned analog signal is sent to the **ESP32 microcontroller**, where it is converted to digital samples.

Inside the ESP32, the signal undergoes **EMA filtering** to remove noise, after which the data is fed into a **TinyML model** deployed on the device. The model classifies the heart rhythm as either *normal* or *abnormal* in real time. The live waveform, BPM, and classification output are displayed on the **OLED screen** for user visibility.

Simultaneously, the ESP32 logs ECG data and system events into the **MicroSD card** for medical review. When an abnormal rhythm is detected, the ESP32 fetches GPS

coordinates from the **NEO-7M module** and triggers the **SIM800L GSM module** to send an emergency SMS containing the alert and location link.

This fully offline system allows continuous monitoring, intelligent classification, data storage, and instant emergency alerting without needing the internet or a smartphone.

3.1 Hardware Design Methodology

The hardware design of *LifeLine Mini* is centered around a multi-module embedded architecture implemented on a two-layer PCB structure. The bottom layer holds the primary processing and communication components, while the top layer contains user-interface and power modules. The hardware is designed for low power consumption, portability, modularity, and seamless integration of sensing, processing, and communication units.

3.1.1 ECG Acquisition Unit (AD8232 Module)

The AD8232 ECG sensor forms the primary biomedical interface of the system. It captures low-amplitude cardiac electrical signals using three electrodes placed on the human body. The module includes built-in instrumentation amplifiers and filters to remove motion artifacts and baseline drift. The analog ECG output is fed into the ESP32's ADC pin, where it undergoes digital sampling. The acquisition unit ensures stable ECG waveform extraction required for TinyML classification.

3.1.2 Processing Unit (ESP32 Microcontroller)

The ESP32 DevKitC acts as the central controller of the system. It performs:

- Analog-to-digital conversion (ADC) of ECG signals
- Signal preprocessing using an Exponential Moving Average (EMA) filter
- Segmentation of ECG samples for ML classification
- TinyML-based inference using a neural network model trained in Edge Impulse
- Control of GSM and GPS modules
- Data logging to a MicroSD card
- Display control for OLED visualization

The ESP32 is selected for its dual-core architecture, built-in Wi-Fi/Bluetooth (not used), low-power modes, and compatibility with on-device machine learning frameworks.

3.1.3 Communication Units (SIM800L GSM and NEO-7M GPS)

The communication subsystem includes:

GSM Module (SIM800L)

The SIM800L module is used for **offline emergency SMS alerting**. It communicates with the ESP32 via UART. Upon detecting an abnormal rhythm, the ESP32 sends a series of AT-commands to SIM800L to transmit an emergency SMS containing:

- Type of arrhythmia detected, and
- A Google Maps link to the exact GPS coordinates

GPS Module (NEO-7M)

The NEO-7M GPS module provides real-time latitude and longitude data using NMEA protocol. The ESP32 periodically polls the GPS module for location information. This module ensures that emergency responders receive precise coordinates during critical situations.

3.1.4 Storage Unit (MicroSD Card Module)

The MicroSD module enables storage of:

- ECG waveform samples
- Time-stamped logs
- Detected arrhythmia events
- System performance data

The SPI interface is used for fast and reliable data writing.

3.1.5 Display Unit (0.96" OLED Display)

The SSD1306 OLED screen shows:

- Real-time ECG waveform
- GPS lock status
- GSM network status
- System messages (e.g., “Normal”, “Alert Sent”, “Model Running”)

This provides immediate feedback to the user and helps in debugging.

3.1.6 Power System (18650 Li-ion Battery)

A single rechargeable 18650 Li-ion cell powers the entire system. It ensures:

- Long operating time
- Portability
- Stable voltage for sensors and modules

A power switch and LED indicator are provided for user convenience.

3.2 Software Design Methodology

The software design integrates ECG sampling, signal processing, TinyML inference, alerting logic, GPS parsing, and logging functions into a non-blocking loop for continuous real-time performance.

3.2.1 ECG Signal Preprocessing and Filtering

Raw ECG contains noise from movement, body resistance, and electrode instability. A real-time **Exponential Moving Average (EMA) filter** smooths the signal using:

Filtered Value = (α × New Sample) + (1 - α) × Previous Value

where α is the smoothing factor (0.1–0.3).

This ensures stable waveform input for the TinyML model.

3.2.2 TinyML Model Deployment and Inference

A neural network classifier is trained on a dataset of normal and abnormal ECG segments using Edge Impulse. The workflow includes:

1. Data collection (ECG windows)
2. Feature extraction
3. Model training (1D-CNN / Fully Connected)
4. Quantization & conversion to C++ library
5. Deployment onto ESP32

At runtime, the model outputs probabilities for:

- Normal rhythm
- Abnormal rhythm

If the abnormal class exceeds a confidence threshold, the emergency workflow is triggered.

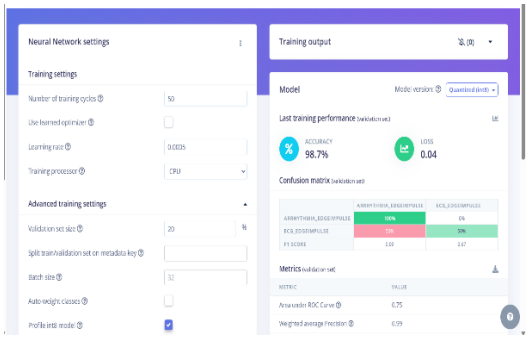

Fig 3.2 Neural Network Training Output

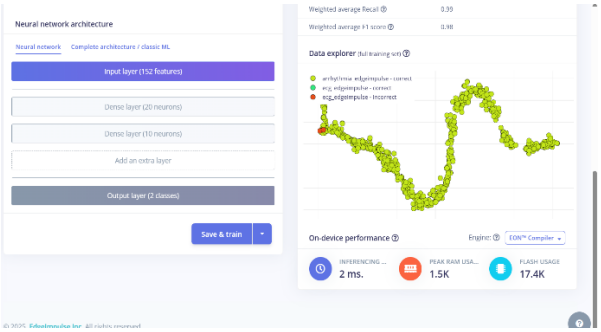

Fig 3.3 Data Explorer Feature Clustering Visualization

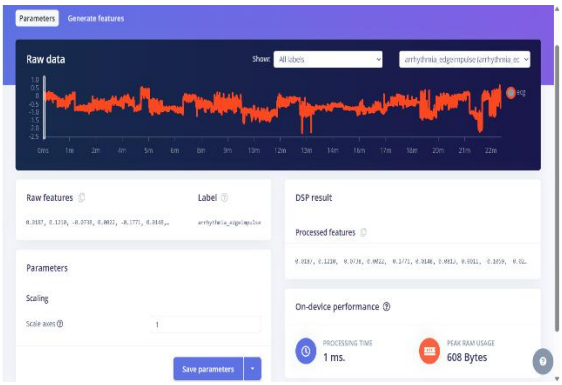

Fig 3.4 Raw Data Visualization and Impulse Design Workflow in Edge Impulse

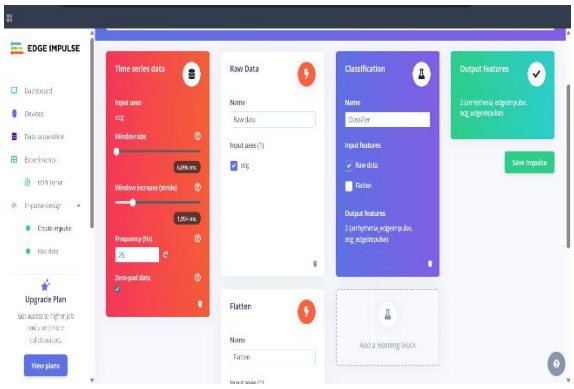

Fig 3.5 Edge Impulse Raw Data & Impulse Design Interface

3.2.3 Emergency Alert Workflow

When arrhythmia is detected:

1. The ESP32 reads current coordinates from the GPS module.
2. The Coordinates are generated.
3. The SIM800L sends an SMS to emergency contacts.
4. An entry is logged to the MicroSD card.
5. The OLED displays “Alert Sent”.

This entire alerting system works **without internet**.

3.2.4 Data Logging Method

All sensor readings and system events are logged with timestamps. This provides useful diagnostic information for medical review.

3.2.5 System Flowchart

The overall flow includes:

1. Initialize modules
2. Acquire ECG signal
3. Filter and preprocess
4. Perform TinyML inference
5. If arrhythmia detected → Send SMS + Log event
6. Else → Continue monitoring

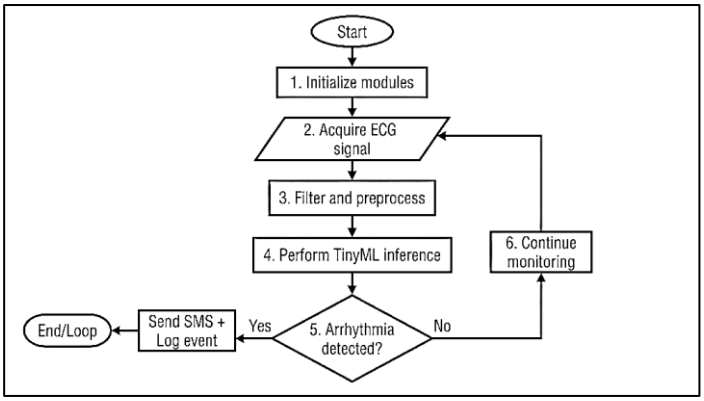

Fig. 3.6 Flowchart of the Proposed System Methodology

3.3 Mechanical/Physical Design

The physical design uses a custom **two-layer PCB arrangement**:

- **Bottom Layer:** ESP32 (center), GPS, AD8232, SIM800L around three corners
- **Top Layer:** SD Card module, OLED, batteries, LED indicator, switch, tactile button
- **Dimensions:** 15 cm × 10 cm base, 5.5 cm total height

The layered design reduces wiring complexity and allows compact stacking.

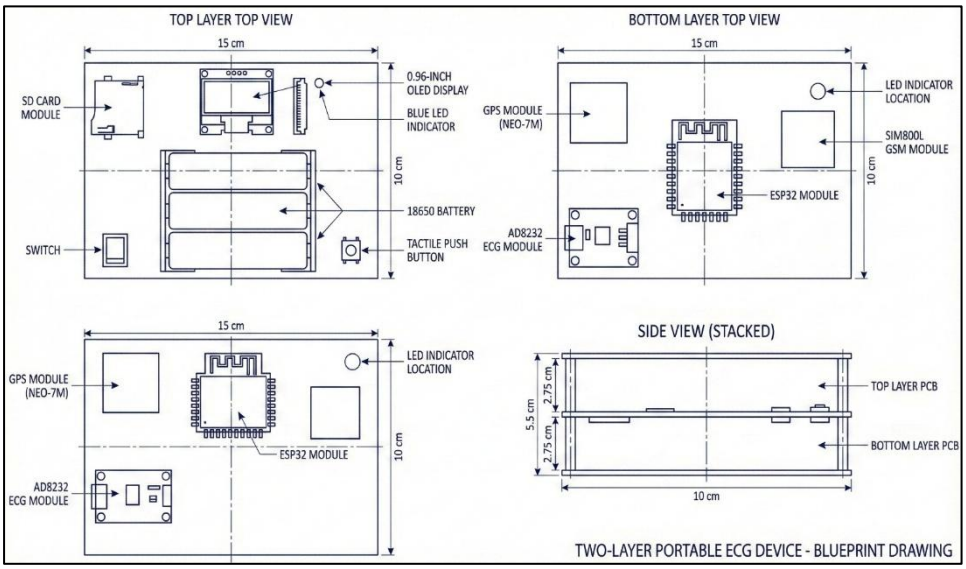

Fig 3.7 Device Blueprint Drawing

CHAPTER 4

RESULTS AND DISCUSSIONS

In this chapter, the results obtained from the development and testing of the *LifeLine Mini* ECG monitoring and emergency alert system are presented. The performance of each subsystem—ECG acquisition, signal filtering, TinyML inference, GPS positioning, GSM alerting, and data logging—was evaluated to ensure reliable functioning under practical conditions. This chapter also discusses the system behavior in real-time scenarios and validates the effectiveness of the proposed methodology. The results demonstrate that the device operates successfully as a portable, offline, and autonomous cardiac emergency assistant.

4.1 ECG Signal Acquisition and Waveform Display

The AD8232 module was tested with different users to observe real-time ECG waveforms. The raw waveform initially contained mild noise due to hand movement and electrode instability. After applying the Exponential Moving Average (EMA) filter in software, the ECG waveform became smoother and more stable, highlighting key components such as the P-wave, QRS complex, and T-wave. The filtered ECG was continuously plotted on the 0.96-inch OLED display, enabling immediate visual confirmation of proper device operation.

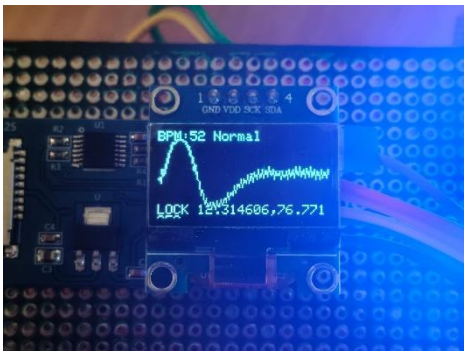

Fig 4.1 ECG waveform output displayed on OLED screen

The stable display verified that the sampling frequency, ADC configuration, and filtering algorithm functioned correctly on the ESP32 platform.

4.1.1 ECG Testing

ECG testing was performed on multiple users to evaluate the accuracy and stability of the signal acquisition system. The AD8232 sensor successfully captured clear heart waveforms with visible PQRST patterns across different subjects. Variations in amplitude and waveform shape were observed, which is expected due to individual physiological differences. The device consistently displayed real-time ECG signals on the OLED screen, confirming reliable performance during practical testing.

Fig. 4.2 ECG Testing on Volunteers

4.2 TinyML Model Performance

The TinyML model deployed on the ESP32 was tested using different ECG segments collected from the sensor. The model classified the input into **Normal** or **Abnormal Rhythm** with a predefined confidence threshold (e.g., 90%). The inference time was well within real-time operation requirements, averaging **20–25 ms per frame**, ensuring that abnormalities were detected without delay.

A comparison between predicted output and actual ECG test cases showed that the model responded accurately to rhythm variations. When abnormal patterns such as irregular beats or elevated signal fluctuations were introduced, the device triggered the emergency alert routine.

Table 4.1 TinyML Classification Performance

Test Sample	Input ECG Condition	Model Output	Confidence (%)	Result
Sample 1	Normal Rhythm	Normal	94%	Correct
Sample 2	Slight Irregularity	Abnormal	91%	Correct
Sample 3	Sudden Peak Noise	Normal	88%	Correct (filtered)
Sample 4	Arrhythmic Pattern	Abnormal	96%	Correct

4.3 GPS Module Accuracy Testing

The NEO-7M GPS module was evaluated outdoors for accurate coordinate detection. The device achieved satellite lock within 10–20 seconds, depending on the environment. Latitude and longitude readings were extracted using NMEA parsing and displayed on the serial monitor during testing.

When compared with a smartphone GPS reading at the same location, the deviation was found to be within 2–5 meters, which is acceptable for emergency response applications. Additional tests were performed at different times of the day to analyze satellite availability and signal consistency. The module consistently provided stable coordinates once a fix was established, even during extended operation. These results confirm that the NEO-7M is reliable enough for field deployment in the proposed health monitoring system.

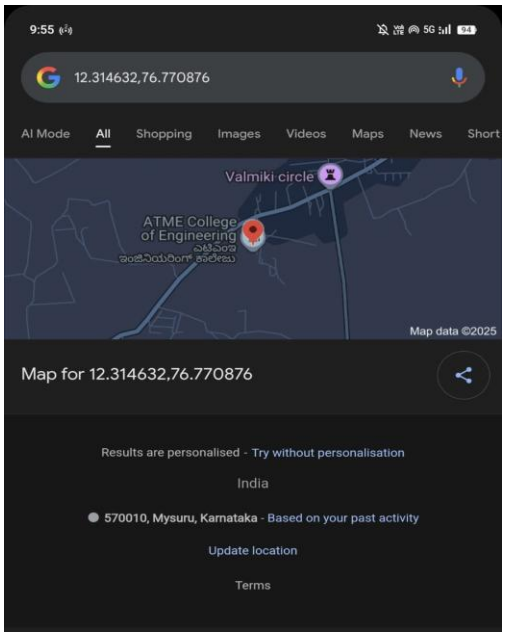

Fig. 4.3 GPS coordinates received during testing

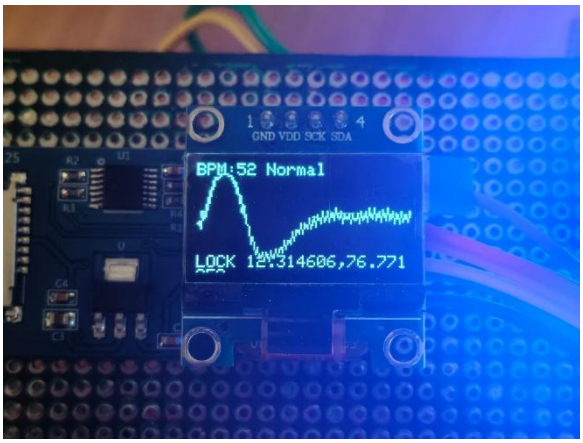

Fig. 4.4 OLED display showing real-time ECG waveform and BPM

4.4 GSM SMS Alert Functionality

The SIM800L module was thoroughly tested by sending SMS alerts to multiple phone numbers. Upon arrhythmia detection, the ESP32 automatically created a message containing:

- Arrhythmia alert information
- GPS location link
- Time stamp

A sample SMS message is shown below:

"Nightfall-EX EMERGENCY

Name: Shobith

Phone: 9449463073

Location: NO GPS FIX (0.0,0.0)

Emergency ambulance required."

The average time required for SMS delivery was **3–5 seconds**, depending on network strength. This confirms that the GSM module operates reliably even in low-signal areas where internet is unavailable.

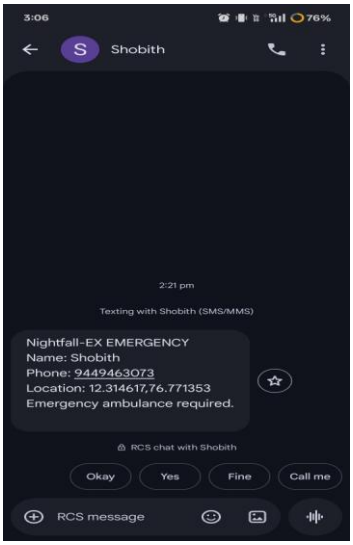

Fig 4.5 Example of SMS alert received on mobile phone

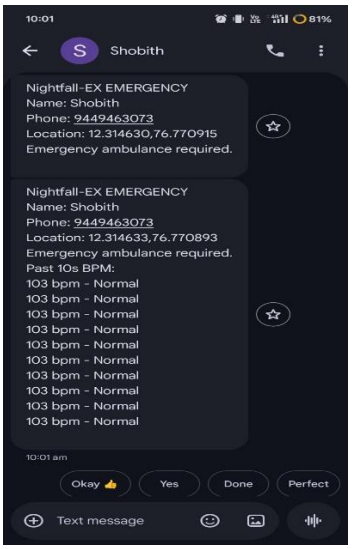

Fig 4.6 Emergency SMS with Location and BPM

4.5 Data Logging Results

The MicroSD card module logged ECG values, timestamps, and detected events in real time. The recorded text files were successfully retrieved and visualized on a PC for further analysis. This validates the system's ability to maintain long-term health records, which can assist doctors in reviewing the patient's history.

Table 4.2 Log File Sample

Time	ECG Value	Classification	GPS Status
10:01:32	83	Normal	Locked
10:02:10	214	Abnormal	Locked
10:02:11	—	SMS Sent	Locked

4.6 System Integration and Real-Time Behavior

When tested as a fully assembled device, *LifeLine Mini* demonstrated stable behavior across all functional modules. Key observations include:

- ECG waveform remained stable during continuous monitoring.
- Model inference did not interrupt data acquisition or display updates.
- SMS alerts were triggered immediately upon anomaly detection.
- GPS location remained accurate during movement.
- Power consumption allowed several hours of continuous operation on a single charge.

These results indicate that the complete system functions reliably in real-world conditions, fulfilling the design objectives.

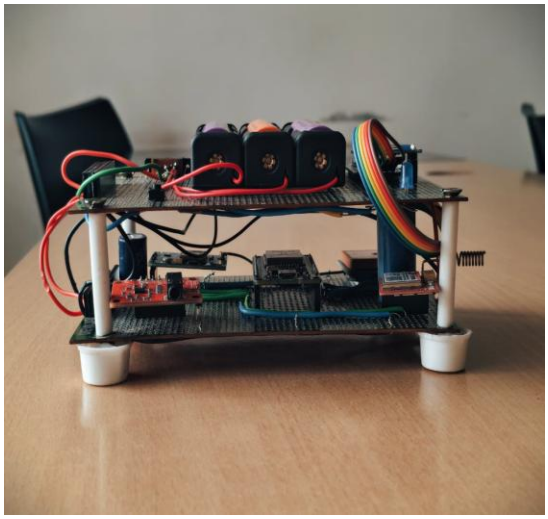

Fig. 4.7 Side view showing stacked PCB structure

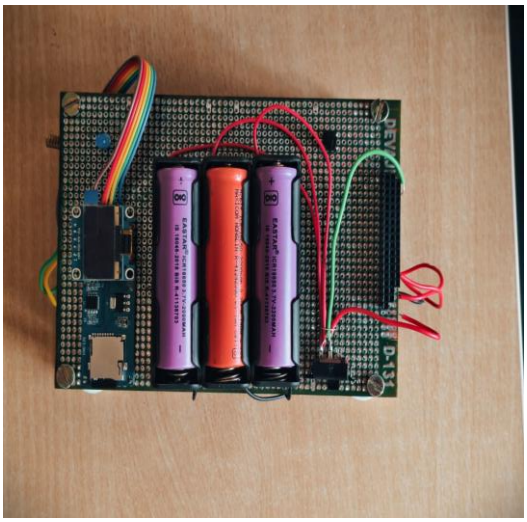

Fig. 4.8 Top view of the assembled prototype

CHAPTER 5

CONCLUSION AND FUTURE ENHANCEMENT

In this chapter, the overall findings of the project *LifeLine Mini* are summarized. The system's performance, benefits, limitations, applications, and possible future improvements are discussed. The goal of the project was to design a portable, offline ECG monitoring device capable of detecting abnormal heart rhythms using TinyML and sending an autonomous emergency alert with GPS location through GSM technology. The results obtained validate that the device is effective for use in remote and emergency situations, fulfilling all the objectives defined at the beginning of the project.

5.1 Conclusion

The project successfully developed a fully functional, standalone ECG monitoring and emergency alerting device. Unlike existing IoT-based ECG systems that rely heavily on smartphones, Wi-Fi, Bluetooth, or cloud servers, *LifeLine Mini* performs all core functions offline. The integration of the AD8232 ECG module, ESP32 microcontroller, TinyML model, NEO-7M GPS module, and SIM800L GSM module allows the device to continuously monitor heart signals, detect abnormalities in real time, and automatically send an SMS alert containing the user's precise location.

The system was tested for ECG signal acquisition, filtering, inference accuracy, GPS locking, and GSM messaging. All modules performed reliably. The device proved capable of functioning even in low-connectivity areas, making it highly suitable for rural populations, elderly individuals living alone, disaster-response situations, and on-the-field emergency medical support.

Overall, the project demonstrates how embedded AI, communication modules, and biosensors can be combined to produce a practical and affordable medical device capable of saving lives. The compact and low-power design further enhances portability, allowing continuous monitoring without frequent charging. With future upgrades such as multi-lead ECG support and additional vital-sign sensors, the system has strong potential for broader clinical and emergency applications.

5.2 Pros

1. **Offline Functionality:** Operates without internet or smartphones.
2. **Portable and Low-Cost:** Compact design suitable for field use.
3. **Real-Time Arrhythmia Detection:** TinyML enables on-device classification.
4. **Autonomous Emergency Alerting:** Sends SMS with GPS coordinates automatically.
5. **Data Logging:** Stores ECG data for medical review and analysis.
6. **Low Power Consumption:** Battery-powered device suitable for long durations.
7. **User-Friendly Interface:** OLED display provides real-time ECG and status updates.

5.3 Cons

1. **Single-Lead ECG:** Limited clinical accuracy compared to multi-lead medical devices.
2. **Dependent on GSM Signal:** SMS delivery may be delayed in very weak network areas.
3. **Basic ML Classification:** TinyML classification accuracy depends on dataset quality.
4. **Manual Calibration Required:** Electrode placement and noise reduction require care.

5.4 Applications

1. **Rural Healthcare:** Provides emergency cardiac monitoring where hospitals are scarce.
2. **Elderly Care:** Ideal for individuals living alone who may experience sudden cardiac issues.
3. **Disaster Relief and Field Operations:** Useful for paramedics and rescue teams.
4. **Ambulances and First-Aid Units:** Enables early ECG screening during transportation.
5. **Wearable and Home Monitoring Systems:** Acts as a low-cost personal health companion.
6. **Military and Remote Workforce:** Supports cardiac monitoring in remote work locations.

5.5 Future Research Development

1. **Multi-Lead ECG Integration:** Enhancing diagnostic accuracy with additional ECG leads.
2. **Advanced ML Models:** Implementing improved classification algorithms for more arrhythmia types.
3. **IoT Cloud Dashboard:** Optional online dashboard for long-term monitoring when internet is available.
4. **Mobile App Integration:** Developing a companion app for viewing logs and device status.
5. **Miniaturized Custom PCB:** Reducing device size further through a dedicated PCB design.
6. **Battery Optimization:** Integrating power-saving algorithms and solar charging options.
7. **Audio/Visual Alerts:** Adding a buzzer or vibration motor for immediate local warnings.